

**MUPP Asset Interlocks and Limits****11/23/2022****Revision 1*****ABSTRACT:***

The purpose of this document is to outline all interlocks and limits of MUPP assets. The range of interlocks and limits is applied to various MUPP valves, agitators, and pumps.

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1.0 Agitator Interlocks

Table 1: Interlock associated with MUPP agitators.

Asset	CM	Interlock ID	Interlock Conditions	On-Delay (s)
AG-101	AG_101	AG-101-INT-01	LT_102.LT_102.PV < 5%	0
AG-321	AG_321	AG-321-INT-01	LT_322.LT_322.PV < 5%	0
AG-661	AG_661	AG-661-INT-01	LT_667.LT_667.PV < 5%	0
AG-121	AG_OFF_121	AG-121-INT-01	LT_122.LT_122.PV < 5%	0
AG-221	AG_OFF_221	AG-221-INT-01	LT_222.LT_222.PV < 5%	0

2.0 Valve Interlocks

Table 2: Interlocks associated with MUPP flow-control-valves and solenoid-valves.

Asset	CM	Interlock ID	Interlock Conditions	On-Delay (s)
FCV-541	FCV_541-R1A	FCV-541-INT-01	IF (LT_541.DACA.PV > 100) THEN Close Valve	150
FCV-622	FCV_622-R1A	FCV-622-INT-01	IF (TT_663.DACA.PV > 80) THEN Close Valve	0
		FCV-622-INT-02	IF (AG_RUN_621.AGRUN621.PVFL == FALSE) THEN Close Valve	0
FCV-642	FCV_642-R1A	FCV-642-INT-01	IF (TT_663.DACA.PV > 80) THEN Close Valve	0
		FCV-642-INT-02	IF (AG_RUN_641.AGRUN641.PVFL == FALSE) THEN Close Valve	0
FCV-662	FCV_662-R1A	FCV-662-INT-01	IF (TT_663.DACA.PV > 80) THEN Close Valve	0
		FCV-662-INT-02	IF (AG_RUN_661.AGRUN661.PVFL == FALSE) THEN Close Valve	0
SV-104	SV-104	SV-104-INT-01	LT_102.LT_102.PV > 102%	0
SV-105	SV-105	SV-105-INT-01	LT_102.LT_102.PV > 102%	0
SV-124	SV-124	SV-124-INT-01	LT_122.LT_122.PV > 102%	0
SV-125	SV-125	SV-125-INT-01	LT_122.LT_122.PV > 102%	0
SV-614	SV-614	SV-614-INT-01	LT_667.LT_667.PV > 102%	0
SV-615	SV-615	SV-615-INT-01	LT_667.LT_667.PV > 102%	0
SV-616	SV-616	SV-616-INT-01	LT_667.LT_667.PV > 102%	0
SV-691	SV-691	SV-691-INT-01	LT_667.LT_667.PV > 102%	0
SV-692	SV-692	SV-692-INT-01	LT_667.LT_667.PV > 102%	0
SV-693	SV-693	SV-693-INT-01	LT_667.LT_667.PV > 102%	0

3.0 Pump Interlocks and Limits

Table 3: Interlocks and limits associated with MUPP pumps.

Asset	CM	Interlock ID	Interlock Conditions			On-Delay (s)
BMP-241	BMP_OFF_241	BMP-241-INT-01			LT_222.LT_222.PV < 5%	0
		BMP-241-INT-02			LT_322.LT_322.PV > 100%	300
		BMP-241-INT-03	AND		BMP_OFF_241.BMP_RUN_241.PVFL == TRUE FT_247.DACA.PV < 5%	5
CRP-341	CRP_OFF_341	CRP-341-INT-01	AND		3WV_368.P2_368.PVFL == TRUE 3WV_368.P1_368.PVFL == FALSE LT_102.DACA.PV > 100	300
					3WV_368.P2_368.PVFL == FALSE 3WV_368.P1_368.PVFL == TRUE LT_122.DACA.PV > 100	300
					FT_347.DACA.PV < 5 CRP_OFF_341.CRP_RUN_341.PVFL == TRUE	5
		CRP-341-INT-02	AND		FT_347.DACA.PV < 5 CRP_OFF_341.CRP_RUN_341.PVFL == TRUE	5
		CRP-341-INT-03	AND		FT_347.DACA.PV < 5 CRP_OFF_341.CRP_RUN_341.PVFL == TRUE	5
		CRP-341-INT-04			LT_322.DACA.PV < 5	0
CUFP-361	CUFP_OFF_361	CUFP-341-INT-01	AND		FT_363.DACA.PV < 5 CUFP_OFF_361.CUFP_RUN_361.PVFL == TRUE	5
					CUFP_OFF_361.CUFP_RUN_361.PVFL == TRUE	5
		CUFP-341-INT-02			LT_222.DACA.PV > 100	300
		CUFP-351-INT-03			LT_322.DACA.PV < 5	0
DP-611	DP_OFF_611	DP-611-INT-01	AND		SV_614.FLAGA.PVFL == FALSE SV_615.FLAGA.PVFL == FALSE SV_616.FLAGA.PVFL == FALSE	0
					SV_614.FLAGA.PVFL == FALSE SV_615.FLAGA.PVFL == FALSE SV_616.FLAGA.PVFL == FALSE	0
					SV_614.FLAGA.PVFL == FALSE SV_615.FLAGA.PVFL == FALSE SV_616.FLAGA.PVFL == FALSE	0
FDP-521	FDP_OFF_521	DP-611-INT-02	AND		FT_613.DACA.PV < 3 DP_OFF_611.DP_RUN_611.PVFL == TRUE	5
		FDP-521-INT-01			LT_501.DACA.PV > 100%	300
		FDP-521-INT-02	AND	OR	FT_523.DACA.PV < 5%	12
					FDP_OFF_521.FDP_RUN_521 == TRUE	
					FDP_REF_521-R1A.FDP_521.PV > 0	
					FDP_REF_521-R1A.AUTO.PVFL == TRUE	
					FDP_REF_521-R1A.AUTO.PVFL == FALSE	
					FDP_REF_521.FDP_521.PV > 0	
					FDP_REF_521.PD_12_6_SWM.PVFL == FALSE	
					FDP_REF_521.PD_12_6_SWM.PVFL == TRUE	
		FDP-521-INT-03			LT_542.DACA.PV < 5%	0
FP-141	FP_OFF_141	FP-141-INT-01	AND		3WV_143.P1_143.PVFL == TRUE 3WV_143.P2_143.PVFL == TRUE LT_122.DACA.PV < 5%	0
					3WV_143.P1_143.PVFL == TRUE 3WV_143.P2_143.PVFL == TRUE LT_122.DACA.PV < 5%	0
					3WV_143.P1_143.PVFL == TRUE 3WV_143.P2_143.PVFL == TRUE LT_122.DACA.PV < 5%	0

Asset	CM	Interlock ID	Interlock Conditions		On-Delay (s)
		FP-141-INT-02	AND	3WV_143.P1_143.PVFL == TRUE	0
				3WV_143.P2_143.PVFL == TRUE	
				LT_102.DACA.PV < 5%	
		FP-141-INT-03		LT_222.DACA.PV > 100%	300
		FP-141-INT-04	AND	FT_148.DACA.PV < 5%	5
				FP_OFF_141.FP_RUN_141.PVFL == TRUE	
LUP-421	LUP_OFF_421	LUP-421-INT-01		LT_422.LA_LL.PV == FALSE	0
		LUP-421-LIM-01		LUP_421 >= 0	-
		LUP-421-LIM-02		LUP_421 <= 30	-
NTP-561	NTP_OFF_561	NTP-561-INT-01		LT_501.DACA.PV < 5%	0
		NTP-561-INT-02	AND	FT_569.DACA.PV < 1%	5
				FT_573.DACA.PV < 1%	
				NTP_OFF_561.NTP_RUN_561.PVFL == TRUE	
PP-681	PP_OFF_681	PP-681-INT-01	AND	[FT_689.DACA.PV + FT_687.DACA.PV] < 5	5
				PP_OFF_681.PP_RUN_681.PVFL == TRUE	
		PP-681-INT-02		LT_667.DACA.PV < 5%	0
		PP-681-LIM-01		PP_681 >= 0	-
		PP-681-LIM-02		PP_681 <= 50	-